



**Department of Electronics and Communication Engineering  
Academic Year 2022 – 2023 (Odd Semester)**

**Degree, Semester & Branch: III Semester B.E. ECE B**

**Course Code & Title: EC3353 Electronic Devices and Circuits**

**Name of the Faculty member (s): Mrs.V.SrIREngaNachiyar**

**Innovative Practice Description**

- **Unit / Topic:** Unit IV / Crystal oscillators
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 14, TLO 15 & TLO 16
- **Activity Chosen:** Jigsaw
- **Justification:**

Jigsaw is a cooperative group activity in which students are interdependent to achieve a common goal. In part one, each group is provided a different problems in Input /Output Quantization Errors. The group members become experts on that particular topic and create a group response.

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

Students said that they felt this activity had improved their understanding of the frequency of oscillation of crystal oscillator. During assessment tests, it is simple to recall the frequency of crystal oscillator.

- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PSO1 | PSO3 |
|-----|-----|-----|-----|-----|-----|------|------|
| CO2 | 1   | 2   | 2   | 2   | 3   | 1    | 1    |

(1 – Low      2 – Moderate      3 – High)

- **PO / PSO mapped:**

| Innovative practice           | PO10                                                                                                                      |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                               | 2                                                                                                                         |
| Justification for correlation | Students discussed with their peers and write the procedures and solving techniques in paper. So, it is mapped at level 3 |

- **Images / Screenshot of the practice:**



**Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

Students stated that they felt that this activity helped them to understand the procedure and steps to calculate the frequency of oscillation of crystal oscillator. It is very useful to recollect the important steps for solving

❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the knowledge about oscillator, function of crystal oscillator and the frequency oscillation. Due to this activity, the students were able to recall the topic and remember the important terms easily.

❖ ***Challenges faced in implementation:***

Students took some time to forming the group. After formed the group, they discussed the topic and present the important terms in paper. I encouraged the students for completing the activity with useful points.

**References:**

- ❖ David A. Bell, "Electronic Devices and Circuits", Oxford Higher Education press, 5 th Edition, 2010.
- ❖ Robert L. Boylestad and Louis Nasheresky, "Electronic Devices and Circuit Theory", 10th Edition, Pearson Education / PHI, 2008.

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23/12/22

**Signature of the Faculty member**

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23/12/22

**HOD/Incharge**

**Department of Electronics and Communication Engineering  
Academic Year 2022 – 2023 (Odd Semester)****Degree, Semester & Branch: III Semester B.E. ECE B****Course Code & Title: EC3353 Electronic Devices and Circuits****Name of the Faculty member (s): Mrs.V.SrirengaNachiyar****Innovative Practice Description**

- **Unit / Topic:** Unit V / DC/DC convertors –Buck-Boost analysis and design
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 19
- **Activity Chosen:** Flipped class
- **Justification:**

The flipped classroom model is based on the idea that traditional teaching is inverted in the sense that what is normally done in class is flipped or switched with that which is normally done by the students out of class. This Activity gives the clear understanding of the particular topic.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

Students were given the task of preparing a presentation on the topic of "DC/DC converters". They described the analysis and design of buck – boost converter". Students were divided into groups and each group will prepared the detailed presentation of DC/DC Converters and deliver their presentation among all the students.

- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO4 | PO10 | PO12 | PSO3 |
|-----|-----|-----|-----|-----|------|------|------|
| CO5 | 2   | 2   | 3   | 1   | 2    | 1    | 2    |

(1 – Low      2 – Moderate      3 – High)

- **PO / PSO mapped:**

| Innovative practice                  | PO10                                                                                                                             | PO12                                                                                                                                                                                          |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | 2                                                                                                                                | 1                                                                                                                                                                                             |
| <b>Justification for correlation</b> | Students will understand the converters and will give clear information during discussion session. Hence it is mapped at level 2 | Students were identify the previous instances in converters where improvements in technology required lifelong learning for practitioners in accordance with changing. Thus level 1 is mapped |

- **Images / Screenshot of the practice:**



**Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

The practise was well-received by students, who said it helped them remember subjects and substance while writing exams. They also noted that because they are presenting in front of all of the students, this activity boosts their confidence.

❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the knowledge of various DC/DC converters. Due to this activity, the students were able to recall the topic and remember the important terms easily.

❖ ***Challenges faced in implementation:***

I have planned this activity for 30 minutes. Students take some time to present the topic. After formed the group, they present the topic with presentation. Students asked doubts regarding the topic before presentation. I've given a brief overview of the Buck Book analysis and encouraged the students to present which it takes another 5 minutes for completing the activity.

**References:**

- ❖ David A. Bell, "Electronic Devices and Circuits", Oxford Higher Education press, 5 th Edition, 2010.
- ❖ Robert L. Boylestad and Louis Nasheresky, "Electronic Devices and Circuit Theory", 10th Edition, Pearson Education / PHI, 2008.

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Department of Electronics and Communication Engineering  
Academic Year: 2022 - 2023 (Odd Semester)

Name of the Faculty member: Mrs.V.SriengaNachiyar  
Course Code & Title : EC3353 Electronic Devices and Circuits  
Semester, Degree & Branch : III B.E ECE B  
Topic : Rectifiers – Full Wave Rectifiers  
Innovative Method : One Minute Paper

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1) Rectifier:<br/>Convert AC input into pulsation DC.</p> <p>2) Efficiency of Full wave rectifier:<br/><math>\eta = 81.06\%</math></p> <p>3) During Positive half cycle <math>Q_1</math> conducts and <math>Q_2</math> does not conduct. Positive half cycle produce at the output.</p> <p>4) During negative half cycle <math>Q_2</math> conducts and <math>Q_1</math> does not conduct. Negative half cycle produce at the output.</p> <p>5) Transformer Utilization Factor (TUF) = 69.1%</p>                                                   | <p>Rectifier:<br/>1. Converting AC to pulsating DC.</p> <p>2. Efficiency of Full wave Rectifier = 81.06%.</p> <p>3. During positive Half Cycle:<br/><math>Q_1</math> - Conduct<br/><math>Q_2</math> - does not conduct<br/>output is positive.</p> <p>4. During negative Half Cycle:<br/><math>Q_1</math> - does not conduct<br/><math>Q_2</math> - Conduct<br/>output is negative.</p> <p>5. TUF: 69.1%</p>                                                                                                                                     |
| <p>1) Rectifier is Rectifier converts AC to pulsating DC.</p> <p>2) Efficiency of full wave rectifier <math>\Rightarrow 81.06\%</math>.</p> <p>3) During positive half cycle <math>\Rightarrow Q_1</math> conducts, <math>Q_2</math> doesn't conduct. It produces the output as positive half cycle.</p> <p>4) During negative half cycle <math>\Rightarrow Q_1</math> doesn't conduct, <math>Q_2</math> conducts. It produces the output as negative half cycle.</p> <p>5) TUF <math>\Rightarrow</math> Transformer Utilization Factor = 69.1%.</p> | <p>1) Rectifiers:-<br/>converting AC to pulsating DC.</p> <p>2) Efficiency of full wave rectifier = 81.06%.</p> <p>3) During positive half cycle,<br/><math>Q_1 \rightarrow</math> conducts<br/><math>Q_2 \rightarrow</math> does not conduct.<br/>output <math>\rightarrow</math> +ve half cycle.</p> <p>4) During Negative half cycle,<br/><math>Q_1 \rightarrow</math> does not conduct.<br/><math>Q_2 \rightarrow</math> conducts.<br/>output <math>\rightarrow</math> -ve half cycle.</p> <p>5) Transformer utilization factor = 69.1%.</p> |

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**Department of Electronics and Communication Engineering**  
**Academic Year: 2022 - 2023 (Odd Semester)**

**Name of the Faculty member:** Mrs.V.SireengaNachiyar  
**Course Code & Title** : EC3353 Electronic Devices and Circuits  
**Semester, Degree & Branch** : III B.E ECE B  
**Topic** : Tuned amplifiers-Double Tuned & its frequency response  
**Innovative Method** : Zero Minute Speech



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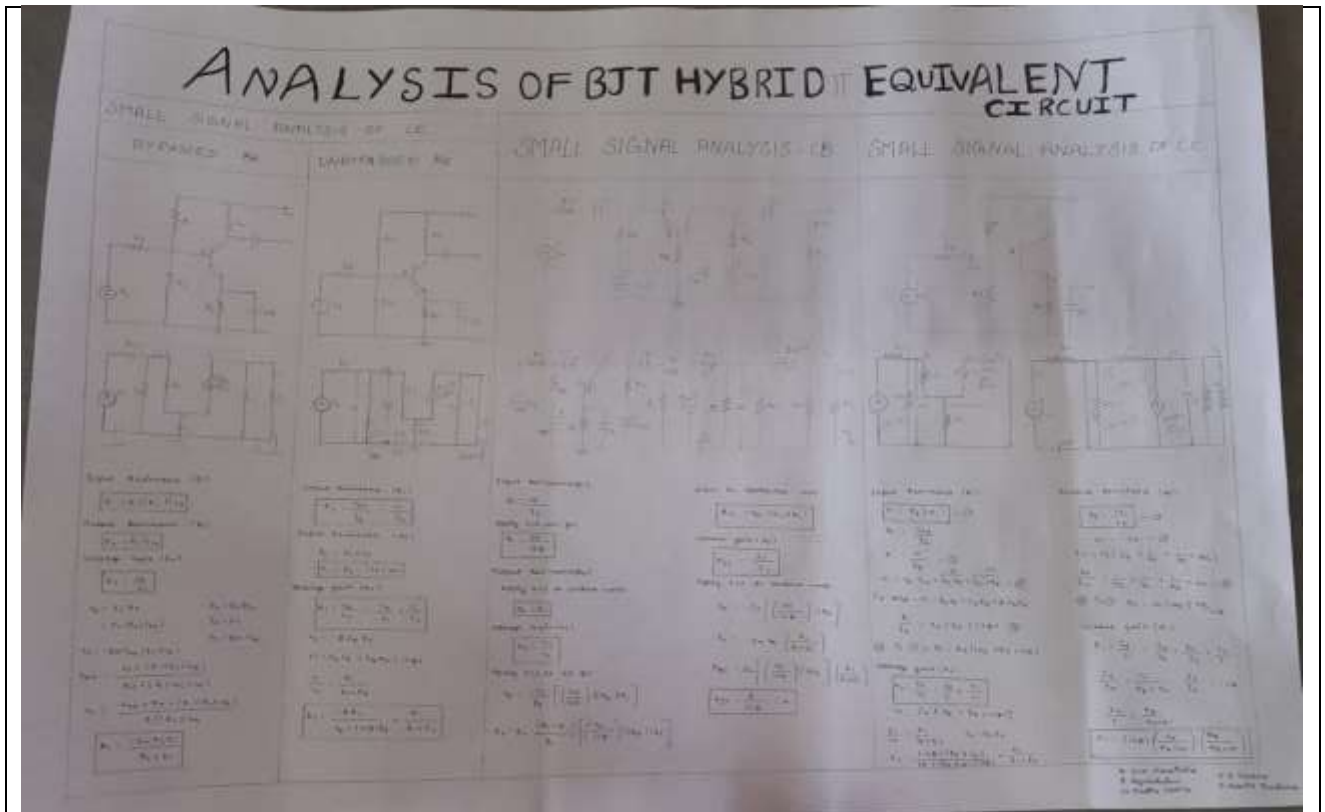
Name of the Faculty member: Mrs.V.SirengaNachiyar

Course Code & Title : EC3353 Electronic Devices and Circuits

Semester, Degree & Branch : III B.E ECE B

Topic : BJT small signal model – Analysis of CC amplifiers- Gain and frequency response

Innovative Method : Chart Preparation



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